

2 (a) Table 2.1 shows descriptions of three types of white blood cell.

Complete Table 2.1 by stating the names of these three types of white blood cell.

Table 2.1

description	name of white blood cell
A large cell that has a bean-shaped (kidney-shaped) nucleus. It can develop into a macrophage.	
A cell that has a large spherical nucleus and little cytoplasm. It responds to non-self antigens.	
A cell that has a lobed nucleus. It is phagocytic.	

[3]

(b) Dromedary camels are classified in the family Camelidae and live in desert habitats of North Africa and Asia. In these hot, dry environments, dromedary camels can lose up to 30% of their body mass from dehydration, causing their blood to become more viscous (thicker).

Fig. 2.1 shows a drawing of red blood cells of a dromedary camel. Fig. 2.2 is a drawing of human red blood cells.

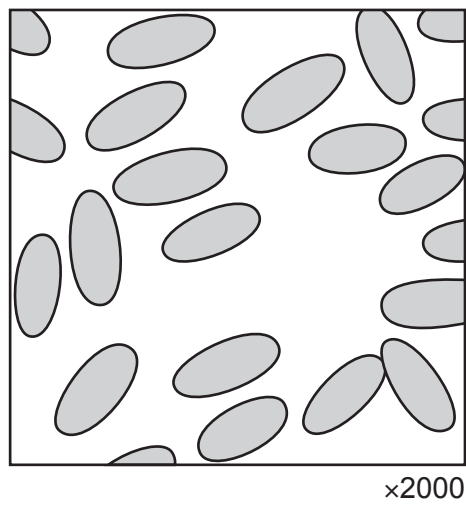


Fig. 2.1

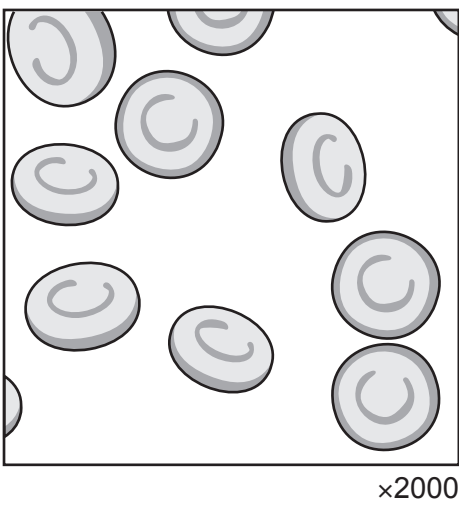


Fig. 2.2

Fig. 2.1 and Fig. 2.2 show differences between the red blood cells of dromedary camels and the red blood cells of humans.

Suggest how these differences adapt dromedary camels for living in hot, dry environments.

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- (c) The llama is also classified in the family Camelidae. Llamas live in mountainous areas of South America, often at altitudes of 3500 m or higher. As the altitude above sea level increases, the air pressure decreases.

The partial pressure of oxygen in the lungs of mammals at 3500 m is 6.4 kPa.

Fig. 2.3 shows the oxygen dissociation curve of adult human haemoglobin and adult llama haemoglobin.

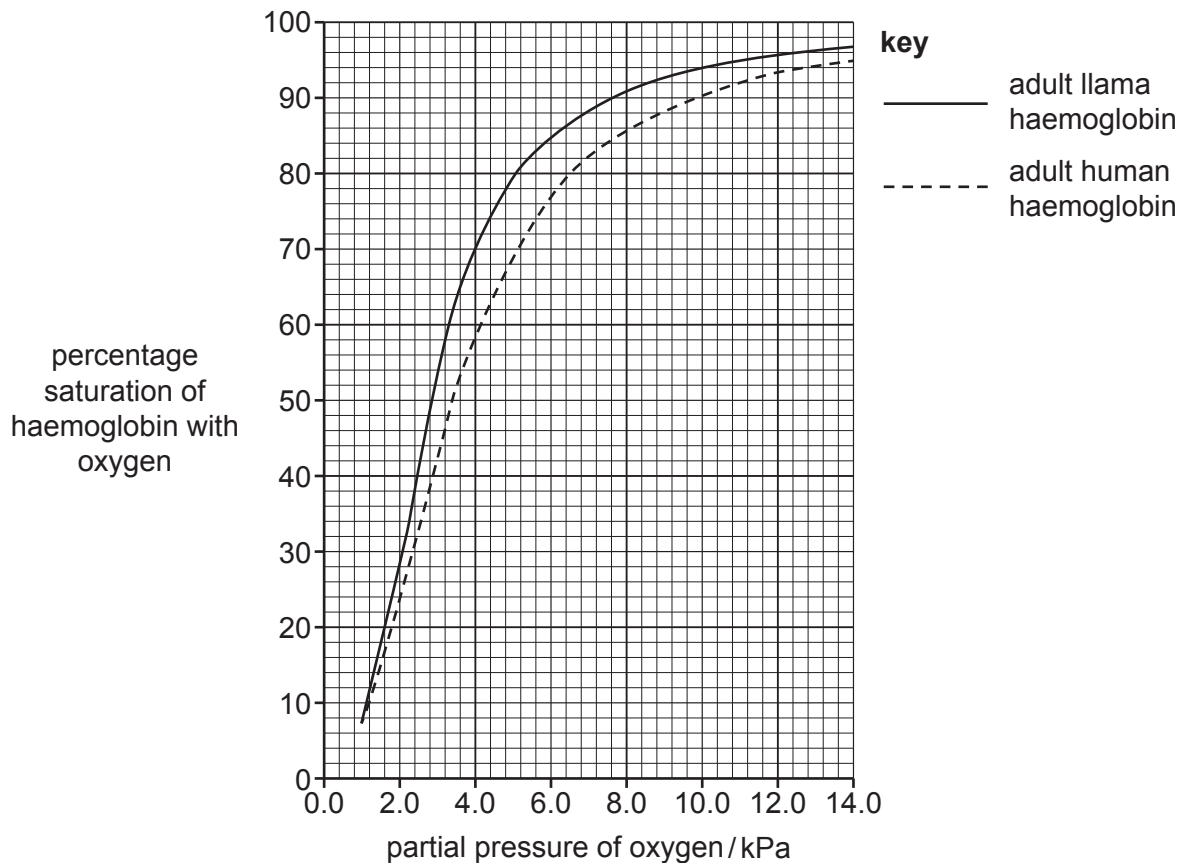


Fig. 2.3

- (i) With reference to Fig. 2.3, explain how the differences between the oxygen dissociation curves for humans and llamas show that llamas are better adapted for living at high altitudes than humans.

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(ii) Sketch a curve on Fig. 2.3 to show the effect of an increased carbon dioxide concentration on the percentage saturation of adult **human** haemoglobin with oxygen. [1]

(iii) Explain the importance of the Bohr shift in metabolically active organs, such as the liver.

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[Total: 13]